

◆ WEEK 2 — SESSION 5

STORAGE LAYER

◆ SESSION OBJECTIVE

By the end of this session, you will understand:

- ◆ how cloud storage works in AWS
 - ◆ what Amazon S3 is
 - ◆ the difference between object storage and block storage
 - ◆ why storage classes matter
 - ◆ how businesses store and manage data in the cloud
 - ◆ and why storage security is critical in cloud environments
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◆ WHAT IS STORAGE?

Simple Definition

Storage means saving and managing digital information.

Applications and systems need storage to:

- ◆ save files
- ◆ store images and videos
- ◆ maintain databases
- ◆ keep backups
- ◆ save logs
- ◆ preserve application data

Without storage, applications cannot remember or retain information.

◆ WHY CLOUD STORAGE MATTERS

Before cloud computing:

- ◆ companies bought physical storage hardware
- ◆ expansion was expensive
- ◆ hardware maintenance was difficult
- ◆ scaling storage took time
- ◆ failures caused operational risks

With AWS cloud storage:

- ◆ storage scales instantly
 - ◆ businesses pay only for usage
 - ◆ infrastructure becomes more reliable
 - ◆ global access becomes easier
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◆ INTRODUCTION TO AMAZON S3

What is Amazon S3?

Amazon S3 is AWS cloud object storage service.

S3 stands for:

Simple Storage Service

It is one of the most widely used cloud storage services in the world.

◆ WHAT S3 IS USED FOR

Businesses use S3 to store:

- ◆ images
 - ◆ videos
 - ◆ backups
 - ◆ application files
 - ◆ documents
 - ◆ logs
 - ◆ static websites
 - ◆ analytics data
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◆ HOW AMAZON S3 WORKS

S3 stores information as objects.

Each object contains:

- ◆ file data
- ◆ metadata
- ◆ unique identifiers

Objects are stored inside containers called:

◆ SIMPLE ANALOGY

Think of an S3 bucket like a cloud warehouse.

Inside the warehouse are digital packages containing files and information.

◆ OBJECT STORAGE VS BLOCK STORAGE

1. OBJECT STORAGE

Object storage stores complete files as objects.

Used for:

- ◆ media files
- ◆ backups
- ◆ cloud applications
- ◆ documents
- ◆ web content

Object storage focuses on:

- ◆ scalability
- ◆ flexibility
- ◆ large-scale storage

Amazon S3 is object storage.

2. BLOCK STORAGE

Block storage breaks data into smaller pieces called blocks.

Used for:

- ◆ operating systems
- ◆ databases
- ◆ virtual machines
- ◆ high-performance applications

Block storage focuses on:

- ◆ speed
 - ◆ performance
 - ◆ low-latency access
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◆ SIMPLE COMPARISON

Object Storage

Like storing complete boxes in a warehouse.

Block Storage

Like cutting information into smaller pieces for faster access and processing.

◆ WHY STORAGE ARCHITECTURE MATTERS

Choosing the wrong storage solution can create:

- ◆ slow performance
- ◆ high operational costs
- ◆ poor scalability
- ◆ inefficient systems

Good cloud engineers choose storage based on:

- ◆ business requirements
 - ◆ performance needs
 - ◆ scalability
 - ◆ cost optimization
 - ◆ security requirements
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◆ S3 STORAGE CLASSES

AWS provides different storage classes for different business needs.

1. S3 STANDARD

Used for:

- ◆ frequently accessed data
 - ◆ active applications
 - ◆ websites
 - ◆ regular workloads
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2. S3 INTELLIGENT-TIERING

Automatically moves data to cheaper storage levels depending on usage.

Helps optimize costs automatically.

3. S3 STANDARD-IA

IA means:

Infrequent Access

Used for:

- ◆ backups
- ◆ files accessed occasionally
- ◆ disaster recovery data

Lower cost than Standard storage.

4. S3 GLACIER

Low-cost archival storage.

Used for:

- ◆ long-term backups
- ◆ archived company records
- ◆ compliance storage

Retrieval is slower but storage is cheaper.

5. S3 GLACIER DEEP ARCHIVE

Very low-cost long-term storage.

Used for:

- ◆ rarely accessed data
 - ◆ long-term compliance archives
 - ◆ historical records
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◆ STORAGE CLASS ANALOGY

Different storage classes are like different storage facilities:

- ◆ expensive fast-access storage
- ◆ lower-cost archive warehouses
- ◆ intelligent automated storage systems

Businesses choose based on:

- ◆ speed needs
 - ◆ access frequency
 - ◆ and budget
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◆ DURABILITY & AVAILABILITY

Amazon S3 is designed for:

- ◆ high durability
- ◆ high availability
- ◆ global scalability

AWS stores data across multiple systems to protect against:

- ◆ hardware failures
 - ◆ storage failures
 - ◆ infrastructure outages
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◆ IMPORTANT UNDERSTANDING

Cloud providers assume hardware will fail eventually.

Cloud architecture is designed to continue operating even during failures.

◆ STORAGE SECURITY RISKS

Poorly configured storage can expose:

- ◆ sensitive company files
- ◆ customer information
- ◆ credentials
- ◆ backups
- ◆ business data

This can lead to:

- ◆ data breaches
 - ◆ compliance failures
 - ◆ financial loss
 - ◆ reputational damage
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◆ STORAGE SECURITY CONTROLS

AWS provides multiple security controls including:

- ◆ bucket permissions
 - ◆ IAM policies
 - ◆ encryption
 - ◆ access restriction
 - ◆ public access blocking
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◆ IMPORTANT SECURITY PRINCIPLE

“Not all data should be publicly accessible.”

Cloud engineers must carefully control storage exposure.

◆ REAL-WORLD BUSINESS USE CASES

Companies use S3 for:

- ◆ application storage
- ◆ media platforms
- ◆ backup systems
- ◆ disaster recovery
- ◆ analytics platforms

- ◆ website hosting
 - ◆ enterprise applications
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◆ PRACTICAL LAB — AMAZON S3

TASK OBJECTIVE

You will build a basic AWS S3 storage environment and understand how cloud storage is managed.

◆ LAB TASKS

- ✓ Create S3 bucket
 - ✓ Upload files
 - ✓ Configure bucket permissions
 - ✓ Test public vs private access
 - ✓ Explore storage classes
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◆ REQUIRED THINKING DURING LAB

Students must evaluate:

1. Security

- ◆ What happens if storage becomes public?
 - ◆ What risks exist with exposed files?
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2. Governance

- ◆ Who should access the storage?
 - ◆ Should all users have full permissions?
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3. Cost Optimization

- ◆ Which storage class is appropriate?
 - ◆ Is expensive storage necessary for old files?
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◆ QUICK RECALL QUESTIONS

1. What is Amazon S3?
 2. What is object storage?
 3. What is the difference between object and block storage?
 4. Why do storage classes exist?
 5. What risks exist with public cloud storage?
 6. Why is storage security important?
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◆ KEY TAKEAWAYS

- ✓ S3 is AWS object storage service
 - ✓ Object and block storage solve different problems
 - ✓ Storage classes optimize cost and performance
 - ✓ Storage security is extremely important
 - ✓ Cloud storage enables scalable modern applications
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◆ FINAL THOUGHT

“Modern cloud systems depend heavily on secure, scalable, and well-governed storage architecture.”